



SEMESTER END EXAMINATIONS – MARCH 2024

Program	: B.E. – CSE (Artificial Intelligence and Machine Learning) / CSE (Cyber Security)	Semester	: III
Course Name	: Data Base Management Systems	Max. Marks	: 100
Course Code	: CI32 / CY32	Duration	: 3 Hrs

Instructions to the Candidates:

- Answer one full question from each unit.

UNIT – I

- Explain with a neat diagram, Component modules of a DBMS and their Interactions. CO1 (08)
 - A university database contains information about professors (identified by social security number SSN) and courses (identified by course id). Professors also have a name, an address and a phone number. Courses have a name and a number of credits. Professors teach courses. For each of the following situations, **Draw an ER diagram** that describes it (assuming no further constraints hold). CO1 (06)
 - Every professor must teach some course.
 - Every professor teaches exactly one course (no more, no less).
 - Every professor teaches exactly one course (no more, no less), and every course must be taught by some professor.
 - Describe the advantages of the DBMS Approach. CO1 (06)
- Explain the Three Schema Architecture with a neat diagram. How does the Three Schema Architecture provide data independence? CO1 (08)
 - Create an ER diagram for a financial banking system.** Identify entities like Account, Customer, Transaction, and Branch. Define relationships and attributes to represent the banking operations. CO1 (06)
 - Differentiate between: CO1 (06)
 - simple and composite attribute.
 - stored and derived attribute.

UNIT – II

- Describe the concept of a weak entity used in data modeling? Define the terms *strong entity type*, *weak entity type*, *identifying relationship type*, and *partial key*. CO2 (06)
 - Describe the following terms as they apply to the relational model of data: *domain*, *attribute*, *n-tuple*, *relation schema*, *relation state*, *degree of a relation*, *relational database schema*, and *relational database state*. CO2 (08)
 - Why are duplicate tuples not allowed in a relation? Explain with suitable example. CO2 (06)
- Discuss the various update operations on relations and the types of integrity constraints that must be checked for each update operation CO2 (06)

- b) Consider the following relations for a database that keeps track of student enrollment in courses and the books adopted for each course: CO2 (08)
- STUDENT(Ssn, Name, Major, Bdate)
 COURSE(Course#, Cname, Dept)
 ENROLL(Ssn, Course#, Quarter, Grade)
 BOOK_ADOPTION(Course#, Quarter, Book_isbn)
 TEXT(Book_isbn, Book_title, Publisher, Author)
- Specify the foreign keys for this schema, stating any assumptions you make.
- c) Discuss the various reasons that lead to the occurrence of NULL values in relations. CO2 (06)

UNIT - III

5. a) Illustrate how are views created and deleted in SQL. Give examples. CO3 (06)
 b) Differentiate between JOIN and NATURAL JOIN operations in SQL. CO3 (06)
 c) Consider the following tables: CO3 (08)
- Branch (branch-name, branch-city, assets)
 Customer (customer-name, customer-street, customer-city)
 loan (loan-number, branch-name, amount)
 Borrower (customer-name, loan-number)
 Account (account-number, branch-name, balance)
 Depositor (customer-name, account-number)
- Write the SQL Queries for the following
- List in alphabetical order all customers who have a loan at 'MSRIT' branch.
 - Find the names of all customers who area address includes the substring "yesvanthpur".
 - List the loan numbers in ascending order of amount.
 - Find the average loan amount at each branch.
6. a) Illustrate the following with syntax and examples: CO3 (08)
- On delete Set NULL and On Update Cascade.
 - IS NULL.
 - Group by and having.
- b) STUDENT (regno: string, name: string, major: string, bdate: date) CO3 (12)
 COURSE (course#:int, cname: string, dept: string)
 ENROLL (regno: string, course#:int, sem: int, marks: int)
 BOOK _ ADOPTION (course# :int, sem: int, book-ISBN: int)
 TEXT (book-ISBN: int, book-title: string, publisher: string, author: string)

Write SQL queries for the following

- List the students name who have enrolled for 2nd sem.
- List no of books adopted in semester '4'.
- List no of books adopted by CSE department.
- List the maximum no of courier adopted by any student.
- List the courses where course name begins with 'a' being held by 'cs' department.
- List department which has more than 2 courses.

UNIT – IV

7. a) Explain any two informal design guidelines for relation schemas. CO4 (05)
 b) Let $R(A,B,C,D,E,G,H)$ be a relation schema with set of functional dependencies: CO4 (06)
 $F = \{AB \rightarrow C, B \rightarrow D, CD \rightarrow E, CE \rightarrow GH, G \rightarrow A\}$.
 With the help of inference rules prove the following
 i. Exhibit a derivation of $AB \rightarrow E$ from F .
 ii. Exhibit a derivation of $GB \rightarrow C$ from F
 c) Consider a relation with schema $R(A, B, C, D)$ and FD's $\{AB \rightarrow C, C \rightarrow D, D \rightarrow A\}$. CO4 (09)
 i. Indicate the nontrivial FD's that can be inferred from the given FD's?
 ii. What are all candidate keys of R ?
 iii. Is this relation in BCNF? If not Decompose R into 2NF, 3NF and BCNF relations.
8. a) Write the algorithm for testing Lossless (nonadditive) Join Property. CO4 (08)
 Consider the schema $R=ABCDE$, subjected to FDs $F = \{A \rightarrow C, B \rightarrow C, C \rightarrow D, DE \rightarrow C, CE \rightarrow A\}$, and the Decomposed relations are $= \{AD, AB, BE, CDE, AE\}$. Check whether the decomposition is satisfying Lossless (nonadditive) Join Property.
 b) Differentiate between 3NF and BCNF. CO4 (07)
 Consider the universal relation $R = \{A, B, C, D, E, F, G, H, I, J\}$ and the set of functional dependencies,
 $F = \{ \{A, B\} \rightarrow \{C\}, \{A\} \rightarrow \{D, E\}, \{B\} \rightarrow \{F\}, \{F\} \rightarrow \{G, H\}, \{D\} \rightarrow \{I, J\} \}$
 What is the key for R ? Decompose R into 2NF and then into 3NF relations.
 c) A set of FD's for the relation $R\{A,B,C,D,E,F\}$ is CO4 (05)
 $AB \rightarrow C, C \rightarrow A, BC \rightarrow D, ACD \rightarrow B, BE \rightarrow C, EC \rightarrow FA, CF \rightarrow BD, D \rightarrow E$. Find a minimum cover for this set of FD'S.

UNIT- V

9. a) Explain ACID properties in Transaction. CO5 (07)
 b) Explain how serializability is used to enforce concurrency control in database System. CO5 (06)
 c) With a neat diagram, write and explain the state transition diagram illustrating the states for transaction execution. CO5 (07)
10. a) What is two phase locking? Will two phase locking result in Serializable schedule? Will two phase locking result in deadlock? Demonstrate with an example. CO5 (10)
 b) Write an Algorithm for testing conflict serializability of a schedule S with an example. CO5 (10)
